System of Linear Equations

Course Overview

- **Textbook**:
  - W. Keith Nicholson (2019), *Linear Algebra with Applications*, Lyrix with Open Texts
  - Gilbert Strang (2016) *Introduction to Linear Algebra*, Wellesley - Cambridge Press, 5th edition
- **Grade:**
  - Test in class (10%)
  - Homework (10%): Exercise on MyOpenMath need to be done on time
  - Attendance (10%) is taken randomly each session
  - Midterm exam (30%)
  - Final exam (40%)

Test, midterm exam and final exam are written examinations

Course Objectives

1. The course provides the student with basic knowledges in linear algebra with introduction to applications to cryptography, computer graphics, image compression . . . , in particular the skill of solving linear systems of equations using Gauss elimination method.
2. **Content**
   - System of linear equations
   - Matrices,operation
   - Determinant, diagonalization
   - Vector space, spanning space, independence and dimension
   - Linear transformation
   - Orthogonalization

Chapter Goals

Systems of linear equations lie at the heart of linear algebra, and this chapter uses them to introduce some of the central concepts of linear algebra in a simple and concrete setting

1. Gaussian Elimination method to solve system of linear equations
2. Introduction to Rank of a matrix via the row-echelon form and relationship between rank of matrix with consistency/inconsistency of a linear system
3. Homogeneous system of linear equations
4. Represent general solutions to a homogeneous system of linear equations as linear combinations of basis solutions - as an introduction to fundamental concepts of linear combination and spaning
5. Introduce an application of linear system in network flow

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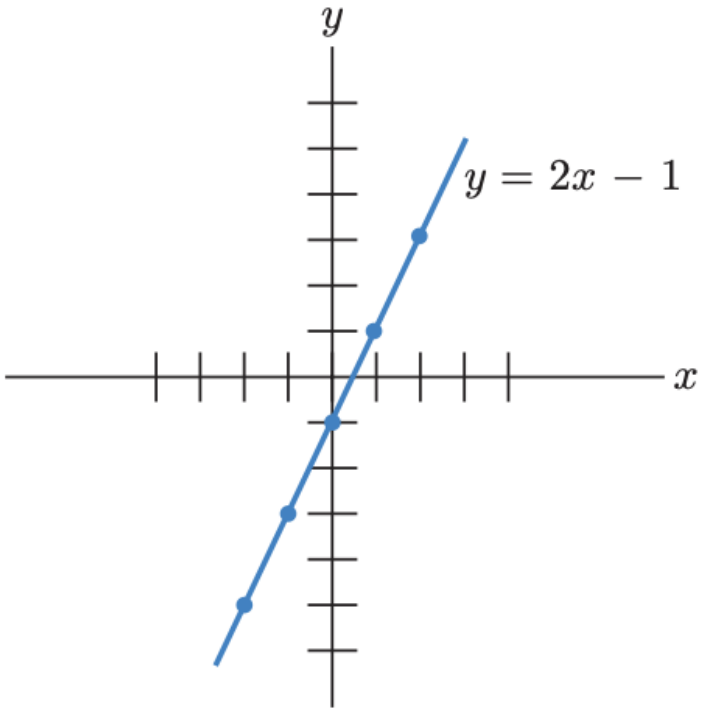
   - Rank of Matrix and Consistency of Linear System
3. Homogeneous System of Equations
   - Homogeneous System
   - Linear combination, general solution and basic solution
4. An Application to Network Flow
5. An Application to Chemical Reactions

Linear equation of two variables

Graph of equation of form

$$ax + by = c$$

is a straight line (if a and b are not both zero), so such an equation is called a linear equation in the variables x and y



Linear equations

An equation of form

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = b$$

is called a **linear equation** in the n variables $x_1, x_2, ..., x_n$

- $a_1, a_2, \ldots, a_n$: **coefficients** of $x_1, x_2, \ldots, x_n$
- b : **constant term**

Each variable in a linear equation occurs to the first power only.

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Each variable in a linear equation occurs to the first power only.

Example

$$2x_1 - 3x_2 + 5x_3 = 7$$

is a linear equation; the coefficients of x_1, x_2, and x_3 are 2, -3, and 5, and the constant term is 7

Solution of a linear equation

Given a linear equation

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = b$$

a sequence $(s_1, s_2, \ldots, s_n)$ **is a solution** of the equation if

$$a_1s_1 + a_2s_2 + \cdots + a_ns_n = b$$

that is, if the equation is satisfied when the substitutions $x_1 = s_1, x_2 = s_2, ..., x_n = s_n$ are made.

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Example

Consider the equation $2x - y = 1$

- $(1, 1)$ is a solution of this equation because

$$2 * 1 - 1 = 1$$

- $(2, 5)$ is not a solution of the equation because

$$2 * 2 - 5 \neq 1$$

System of linear equations

Definition

- A finite collection of linear equations in the variables $x_1, x_2, ..., x_n$ is called a **system of linear equations** in these variables
- A sequence of numbers is called **a solution to a system of equations** if it is a solution to every equation in the system.

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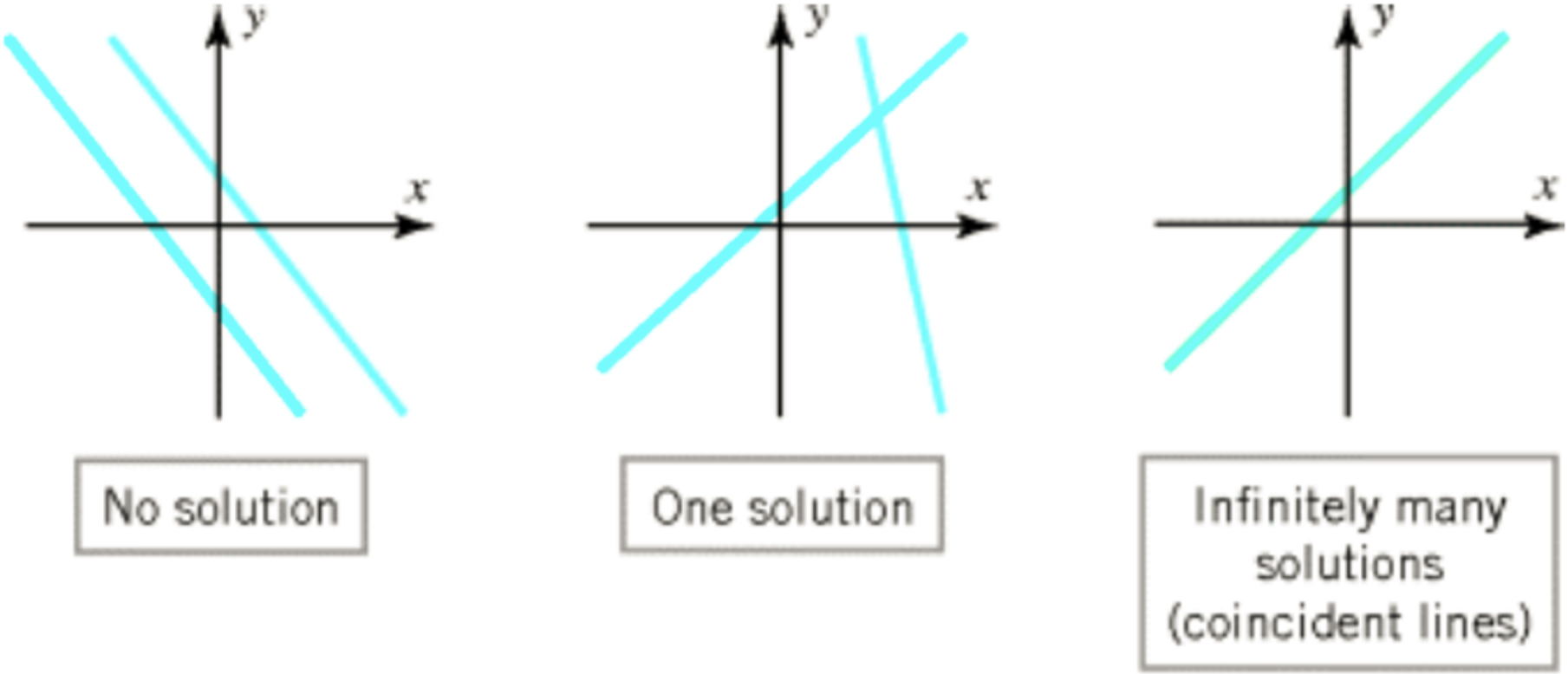
Example

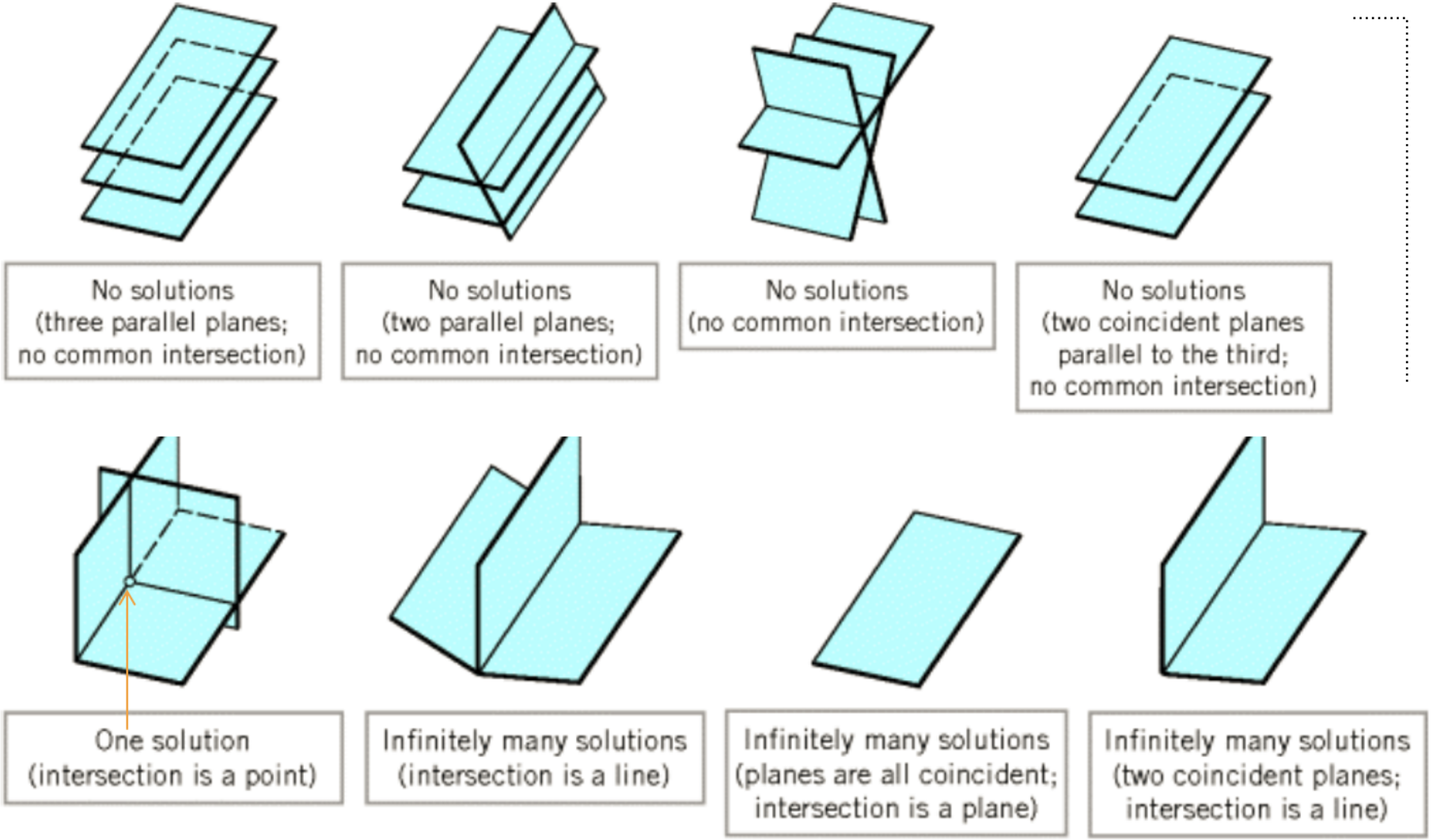
$(x = -2, y = 5, z = 0)$ and $(x = 0, y = 4, z = -1)$ are both solutions to the system

$$\begin{cases} x + y + \ \ z = 3 \\ 2x + y + 3z = 1 \end{cases}$$

System of $ax + by = c$



System of $ax + by + cz = d$



Consistent vs Inconsistent

- A system that has **no solution** is called **inconsistent**
- A system **with at least one solution** is called **consistent**
  - unique solution
  - infinitely many solutions

Example

The system equation

$$\begin{cases} x + 2y = 1 \\ x + 2y = 3 \end{cases}$$

has no solution $\Longrightarrow$ **inconsistent**

Example

For arbitrary values of s and t,

$$\begin{aligned} x_1 &= t - s + 1 \\ x_2 &= t + s + 2 \\ x_3 &= s \\ x_4 &= t \end{aligned}$$

is a solution to the system

$$\begin{cases} x_1 - 2x_2 + 3x_3 + x_4 = -3 \\ 2x_1 - \ \ x_2 + 3x_3 - x_4 = 0 \end{cases}$$

$\Longrightarrow$ has infinitely many solution $\Longrightarrow$ **consistent**

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The quantities s and t in are called **parameters**, and the set of solutions , described in this way, that is $(t - s + 1, t + s + 2, s, t)$, is said to be given in **parametric form** and is called the **general solution** to the system.

Example

Describe all solutions to $3x - y + 2z = 6$ in parametric form.

Example

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Solution

Solving the equation for y in terms of x and z, we get $y = 3x + 2z - 6$. If s and t are arbitrary then, setting $x = s, z = t$, we get solutions

$$\begin{aligned} x &= s \\ y &= 3s + 2t - 6 \\ z &= t \end{aligned}$$

where s and t are arbitrary

Algebraic Method to Solve Linear System

- **Two systems** are said to be **equivalent** if they have the **same set of solutions**.
- A system is **solved by writing a series of systems, one after the other, each equivalent to the previous system**.
- Each of these systems has the same set of solutions as the original one; the aim is to end up with a system that is easy to solve.
- Each system in the series is obtained from the preceding system by a simple manipulation chosen so that it does not change the set of solutions.

Solve system of linear equation with eliminating

Consider the system

$$x + 2y + 3z = 6 \tag{1}$$
$$2x - 3y + 2z = 14 \tag{2}$$
$$3x + y - z = -2 \tag{3}$$

To eliminate x in Eq (2)-(3), we take Eq (2) - 2Eq (1) and Eq (3) - 3Eq (1) and obtain a system of 2 variables y and z

$$-7y - 4z = 2 \tag{4}$$
$$-5y - 10z = -20 \tag{5}$$

Divide Eq (5) by -5

$$-7y - 4z = 2 \tag{6}$$
$$y + 2z = 4 \tag{7}$$

Interchanging equations

$$y + 2z = 4 \tag{8}$$
$$-7y - 4z = 2 \tag{9}$$

Eliminate y in Eq (9) by Eq (8) + 7Eq (9)

$$10z = 30 \text{ or } z = 3$$

After eliminating process, we obtain

$$x + 2y + 3z = 6 \tag{10}$$
$$y + 2z = 4 \tag{11}$$
$$z = 3 \tag{12}$$

Substitute $z = 3$ to Eq (11) to get $y = -2$ and then substitute value of y and z to Eq (11) to obtain $x = 1$. Hence the solution is $(x, y, z) = (1, -2, 3)$. Such process to find the solution is called **back substitution procedure**

Name of variables does not matter

Repeat all the above steps to solve the system equations

$$\begin{cases} x_1 + 2x_2 + 3x_3 = 6 \\ 2x_1 - 3x_2 + 2x_3 = 14 \\ 3x_1 + \ \ x_2 - \ \ x_3 = -2 \end{cases}$$

We only need to replace x, y, z by x_1, x_2, x_3 and keep the same coefficients and constant terms in the eliminating process.

Simplify Computation of Algebraic Method With Matrix

A rectangular array of numbers is called a **matrix**

$$\begin{aligned} 3x_1 + 2x_2 - x_3 + \quad x_4 &= -1 \\ 2x_1 \qquad\quad - x_3 + 2x_4 &= 0 \\ 3x_1 + \ x_2 + 2x_3 + 5x_4 &= 2 \end{aligned} \longrightarrow \left[\begin{array}{cccc|c} 3 & 2 & -1 & 1 & -1 \\ 2 & 0 & -1 & 2 & 0 \\ 3 & 1 & 2 & 5 & 2 \end{array}\right]$$

augmented matrix

coefficient matrix

$$\begin{bmatrix} 3 & 2 & -1 & 1 \\ 2 & 0 & -1 & 2 \\ 3 & 1 & 2 & 5 \end{bmatrix}$$

constant matrix

$$\begin{bmatrix} -1 \\ 0 \\ 2 \end{bmatrix}$$

In **augmented matrix**, each row of the matrix consists of the **coefficients** of the variables (in order) from the corresponding equation, together with the **constant term**

Example

Solve

$$\begin{cases} x - y = 1 \\ x + 2y = -2 \end{cases}$$

augmented matrix $$\left\langle \begin{array}{cc|c} 1 & -1 & 1 \\ 1 & 2 & -2 \end{array} \right\rangle$$

Solution

Apply algebraic method

$$\begin{cases} x - y = 1 \\ x + 2y = -2 \end{cases} \Leftrightarrow \begin{cases} x - y = 1 \\ 0x + 3y = -3 \end{cases} \Leftrightarrow \begin{cases} x - y = 1 \\ 0x + y = -1 \end{cases} \Leftrightarrow \begin{cases} x + 0y = 0 \\ 0x + y = -1 \end{cases}$$

Solution

Apply algebraic method

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The corresponding augmented matrix

$$\left\langle \begin{matrix} 1 & -1 \\ 1 & 2 \end{matrix} \middle| \begin{matrix} 1 \\ -2 \end{matrix} \right\rangle \rightarrow \left\langle \begin{matrix} 1 & -1 \\ 0 & 3 \end{matrix} \middle| \begin{matrix} 1 \\ -3 \end{matrix} \right\rangle \rightarrow \left\langle \begin{matrix} 1 & -1 \\ 0 & 1 \end{matrix} \middle| \begin{matrix} 1 \\ -1 \end{matrix} \right\rangle \rightarrow \left\langle \begin{matrix} 1 & 0 \\ 0 & 1 \end{matrix} \middle| \begin{matrix} 0 \\ -1 \end{matrix} \right\rangle$$

Solution

Apply algebraic method

$$\begin{cases} x - y = 1 \\ x + 2y = -2 \end{cases} \Leftrightarrow \begin{cases} x - y = 1 \\ 0x + 3y = -3 \end{cases} \Leftrightarrow \begin{cases} x - y = 1 \\ 0x + y = -1 \end{cases} \Leftrightarrow \begin{cases} x + 0y = 0 \\ 0x + y = -1 \end{cases}$$

The corresponding augmented matrix

$$\left\langle \begin{array}{cc|c} 1 & -1 & 1 \\ 1 & 2 & -2 \end{array} \right\rangle \rightarrow \left\langle \begin{array}{cc|c} 1 & -1 & 1 \\ 0 & 3 & -3 \end{array} \right\rangle \rightarrow \left\langle \begin{array}{cc|c} 1 & -1 & 1 \\ 0 & 1 & -1 \end{array} \right\rangle \rightarrow \left\langle \begin{array}{cc|c} 1 & 0 & 0 \\ 0 & 1 & -1 \end{array} \right\rangle$$

Solution $(x, y) = (0, -1)$

Elementary operations

The following operations, called **elementary operations**, can routinely be performed on systems of linear equations **to produce equivalent systems**.

1. (Interchange) Interchange two equations.
2. (Scaling) Multiply one equation by a nonzero number.
3. (Replacement) Add a multiple of one equation to a different equation.

Use Elementary Row Operations

The following are called elementary row operations on a matrix.

1. Interchange two rows.
2. Multiply one row by a nonzero number.
3. Add a multiple of one row to a different row.

To Produce a Matrix Of the Form

$$\left[\begin{array}{cc|c} 1 & 0 & * \\ 0 & 1 & * \end{array}\right]$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & * \\ 0 & 1 & 0 & * \\ 0 & 0 & 1 & * \end{array}\right]$$

Use Elementary Row Operations

The following are called elementary row operations on a matrix.

1. Interchange two rows.
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To Produce a Matrix Of the Form

$$\left[\begin{array}{cc|c} 1 & 0 & * \\ 0 & 1 & * \end{array}\right]$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & * \\ 0 & 1 & 0 & * \\ 0 & 0 & 1 & * \end{array}\right]$$

But this does not always happen.

Example

Find all solutions to the following system of equations.

$$3x + 4y + z = 1$$
$$2x + 3y = 0$$
$$4x + 3y - z = -2$$

Solution

Reduction the augmented matrix

$$\left[\begin{array}{ccc|c} 3 & 4 & 1 & 1 \\ 2 & 3 & 0 & 0 \\ 4 & 3 & -1 & -2 \end{array}\right] \xrightarrow[\frac{1}{2}r_2]{r_1 \leftrightarrow r_2} \left[\begin{array}{ccc|c} 1 & \frac{3}{2} & 0 & 0 \\ 3 & 4 & 1 & 1 \\ 4 & 3 & -1 & -2 \end{array}\right]$$ create a 1 in the upper left corner

$$\xrightarrow[r_2 - 3r_1]{r_3 - 4r_1} \left[\begin{array}{ccc|c} 1 & \frac{3}{2} & 0 & 0 \\ 0 & -\frac{1}{2} & 1 & 1 \\ 0 & -3 & -1 & -2 \end{array}\right]$$ creat zeros in other position in col 1

$$\xrightarrow[-2r_2]{} \left[\begin{array}{ccc|c} 1 & \frac{3}{2} & 0 & 0 \\ 0 & 1 & -2 & -2 \\ 0 & -3 & -1 & -2 \end{array}\right]$$ create 1 leading row 2

$$\xrightarrow{r_3 + 3r_2} \left[\begin{array}{ccc|c} 1 & \frac{3}{2} & 0 & 0 \\ 0 & 1 & -2 & -2 \\ 0 & 0 & -7 & -8 \end{array}\right]$$ create 0 in position below 1 leading in col 2

Create 1 leading in row 2

$$\xrightarrow{-\frac{1}{7}r_3} \left[\begin{array}{ccc|c} 1 & \frac{3}{2} & 0 & 0 \\ 0 & 1 & -2 & -2 \\ 0 & 0 & 1 & \frac{8}{7} \end{array}\right]$$

The corresponding system of equations

$$\begin{cases} x + \frac{3}{2}y \qquad\quad = 0 \\ \qquad\quad y - 2z = -2 \\ \qquad\qquad\quad z = \frac{8}{7} \end{cases}$$

which give the unique solution $(x = -\frac{3}{7}, y = \frac{2}{7}, z = \frac{8}{7})$.

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Row-Echelon Form

$$\begin{bmatrix} 0 & 1 & * & * & * & * & * \\ 0 & 0 & 0 & 1 & * & * & * \\ 0 & 0 & 0 & 0 & 1 & * & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

- All the zero rows are at the bottom
- The first nonzero entry from the left in each nonzero row is a 1, called the **leading 1** for that row
- Each leading 1 is **to the right** of all leading 1's in the rows above it

Stair form of row echelon matrix

leading ones

$$\begin{bmatrix} 0 & 1 & * & * & * & * & * \\ 0 & 0 & 0 & 1 & * & * & * \\ 0 & 0 & 0 & 0 & 1 & * & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

The leading 1s proceed "down and to the right" through the matrix. All entries below and to the left of them are zero.

Which is a row-echelon matrix?

$$\begin{bmatrix} 1 & * & * \\ 0 & 0 & 1 \end{bmatrix} \qquad \begin{bmatrix} 1 & * & * \\ 0 & 2 & * \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & * & * & * \\ 0 & 1 & * & * \\ 0 & 1 & * & * \end{bmatrix} \qquad \begin{bmatrix} 0 & 1 & * & * \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \qquad \begin{bmatrix} 1 & * & * \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Reduced row-echelon matrix

- It is a row-echelon matrix
- Each leading 1 is the only nonzero entry in its column

$$\begin{bmatrix} 1 & 0 & * & 0 \\ 0 & 1 & * & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Which is a reduced row- echelon matrix?

$$\begin{bmatrix} 1 & * & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\cancel{\begin{bmatrix} 1 & * & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}}$$

$$\begin{bmatrix} 1 & 0 & * & 0 \\ 0 & 1 & * & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & * & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\cancel{\begin{bmatrix} 1 & * & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}}$$

Carry a matrix to (reduced) row - echelon form with Gaussian Algorithm

Every matrix can be brought to (reduced) row-echelon form by a series of elementary row operations

Carry a matrix to (reduced) row - echelon form with Gaussian Algorithm

Every matrix can be brought to (reduced) row-echelon form by a series of elementary row operations

Gaussian Algorithm

Step 1 If the matrix consists entirely of zeros, stop

Step 2 Otherwise, find the first column from the left containing a nonzero entry (call it **a**), and move the row containing that entry to the top position.

Step 3 Now multiply the new top row by $1/\mathbf{a}$ to create a leading 1.

Step 4 By subtracting multiples of that row from rows below it, make each entry below the leading 1 zero.

Step 5 Repeat steps 1–4 on the matrix consisting of the remaining rows.

Ilustrate a part of Gaussian Algorithm

$$step\ 1: \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow stop$$

leading one

$$\begin{bmatrix} 0 & 0 & 1 & 3 \\ 2 & 2 & -1 & 0 \\ 0 & 3 & 0 & 6 \\ 4 & 7 & 5 & -1 \end{bmatrix} \xrightarrow{r_1 \leftrightarrow r_2} \begin{bmatrix} 2 & 2 & -1 & 0 \\ 0 & 0 & 1 & 3 \\ 0 & 3 & 0 & 6 \\ 4 & 7 & 5 & -1 \end{bmatrix} \xrightarrow{\frac{1}{2}r_1} \begin{bmatrix} 1 & 1 & -\frac{1}{2} & 0 \\ 0 & 0 & 1 & 3 \\ 0 & 3 & 0 & 6 \\ 4 & 7 & 5 & -1 \end{bmatrix} \xrightarrow{-4r_1+r_4} \begin{bmatrix} 1 & 1 & -\frac{1}{2} & 0 \\ 0 & 0 & 1 & 3 \\ 0 & 3 & 0 & 6 \\ 0 & 3 & 7 & -1 \end{bmatrix}$$

step 2 **step 3** **step 4** **step 5**

Example

Carry the matrix

$$\begin{bmatrix} 2 & 6 & -2 & 2 \\ -2 & -3 & 11 & 4 \\ 3 & 11 & 3 & 0 \end{bmatrix}$$

to

1. row - echelon form
2. reduced row - echelon form

Solution - Carry to echelon - form

$$\begin{bmatrix} 2 & 6 & -2 & 2 \\ -2 & -3 & 11 & 4 \\ 3 & 11 & 3 & 0 \end{bmatrix} \xrightarrow{\frac{1}{2}r_1} \begin{bmatrix} 1 & 3 & -1 & 1 \\ -2 & -3 & 11 & 4 \\ 3 & 11 & 3 & 0 \end{bmatrix} \xrightarrow[-3r_1+r_3]{2r_1+r_2} \begin{bmatrix} 1 & 3 & -1 & 1 \\ 0 & 3 & 9 & 6 \\ 0 & 2 & 6 & -3 \end{bmatrix}$$

$$\xrightarrow{\frac{1}{3}r_2} \begin{bmatrix} 1 & 3 & -1 & 1 \\ 0 & 1 & 3 & 2 \\ 0 & 2 & 6 & -3 \end{bmatrix} \xrightarrow{-2r_2+r_3} \begin{bmatrix} 1 & 3 & -1 & 1 \\ 0 & 1 & 3 & 2 \\ 0 & 0 & 0 & -7 \end{bmatrix} \xrightarrow{-\frac{1}{7}r_3} \begin{bmatrix} 1 & 3 & -1 & 1 \\ 0 & 1 & 3 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

row-echelon matrix

Continue to carry to reduced echelon - form

$$\begin{bmatrix} 1 & 3 & -1 & 1 \\ 0 & 1 & 3 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix} \xrightarrow[-r_3+r_1]{-2r_3+r_2} \begin{bmatrix} 1 & 3 & -1 & 0 \\ 0 & 1 & 3 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\xrightarrow{-3r_2+r_1} \begin{bmatrix} 1 & 0 & -10 & 0 \\ 0 & 1 & 3 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

reduced row-echelon matrix

Example

Solve the system equations

$$\begin{cases} 3x + y - \ \ 4z = -1 \\ \ \ x \qquad + 10z = 5 \\ 4x + y + \ \ 6z = 1 \end{cases}$$

Solution

Augmented matrix

$$\left[\begin{array}{ccc|c} 3 & 1 & -4 & -1 \\ 1 & 0 & 10 & 5 \\ 4 & 1 & 6 & 1 \end{array}\right] \xrightarrow[r_1 \leftrightarrow r_2]{\text{create first leading 1}} \left[\begin{array}{ccc|c} 1 & 0 & 10 & 5 \\ 3 & 1 & -4 & -1 \\ 4 & 1 & 6 & 1 \end{array}\right]$$

$$\xrightarrow[r_2 - 3r_1]{r_3 - 4r_1} \left[\begin{array}{ccc|c} 1 & 0 & 10 & 5 \\ 0 & 1 & -34 & -16 \\ 0 & 1 & -34 & -19 \end{array}\right] \xrightarrow[r_3 - r_2]{} \left[\begin{array}{ccc|c} 1 & 0 & 10 & 5 \\ 0 & 1 & -34 & -16 \\ 0 & 0 & 0 & -3 \end{array}\right]$$

The following reduced system of equations

$$\begin{cases} x + 0y + 10z = 5 \\ 0x + \ \ y - 34z = -16 \\ 0x + 0y + \ \ 0z = -3 \end{cases}$$

is equivalent to the original system.

Solution

Augmented matrix

$$\left[\begin{array}{ccc|c} 3 & 1 & -4 & -1 \\ 1 & 0 & 10 & 5 \\ 4 & 1 & 6 & 1 \end{array}\right] \xrightarrow[r_1 \leftrightarrow r_2]{\text{create first leading 1}} \left[\begin{array}{ccc|c} 1 & 0 & 10 & 5 \\ 3 & 1 & -4 & -1 \\ 4 & 1 & 6 & 1 \end{array}\right]$$

$$\xrightarrow[r_2 - 3r_1]{r_3 - 4r_1} \left[\begin{array}{ccc|c} 1 & 0 & 10 & 5 \\ 0 & 1 & -34 & -16 \\ 0 & 1 & -34 & -19 \end{array}\right] \xrightarrow[r_3 - r_2]{} \left[\begin{array}{ccc|c} 1 & 0 & 10 & 5 \\ 0 & 1 & -34 & -16 \\ 0 & 0 & 0 & -3 \end{array}\right]$$

The following reduced system of equations

$$\begin{cases} x + 0y + 10z = 5 \\ 0x + \ \ y - 34z = -16 \\ 0x + 0y + \ \ 0z = -3 \end{cases}$$

is equivalent to the original system. The last system has no solution hence the origin system has no solution.

Example

Solve the following system of equations.

$$\begin{cases} x_1 - 2x_2 - x_3 + 3x_4 = 1 \\ 2x_1 - 4x_2 + x_3 = 5 \\ x_1 - 2x_2 + 2x_3 - 3x_4 = 4 \end{cases}$$

Solution

The augmented matrix

$$\left[\begin{array}{cccc|c} 1 & -2 & -1 & 3 & 1 \\ 2 & -4 & 1 & 0 & 5 \\ 1 & -2 & 2 & -3 & 4 \end{array}\right] \xrightarrow[r_2-2r_1]{r_3-r_1} \left[\begin{array}{cccc|c} 1 & -2 & -1 & 3 & 1 \\ 0 & 0 & 3 & -6 & 3 \\ 0 & 0 & 3 & -6 & 3 \end{array}\right]$$

$$\xrightarrow[\frac{1}{3}r_2]{r_3-r_2} \left[\begin{array}{cccc|c} 1 & -2 & -1 & 3 & 1 \\ 0 & 0 & 1 & -2 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right] \xrightarrow[r_1+r_2]{} \left[\begin{array}{cccc|c} 1 & -2 & 0 & 1 & 2 \\ 0 & 0 & 1 & -2 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

Solution

The augmented matrix

$$\left[\begin{array}{cccc|c} 1 & -2 & -1 & 3 & 1 \\ 2 & -4 & 1 & 0 & 5 \\ 1 & -2 & 2 & -3 & 4 \end{array}\right] \xrightarrow[r_2-2r_1]{r_3-r_1} \left[\begin{array}{cccc|c} 1 & -2 & -1 & 3 & 1 \\ 0 & 0 & 3 & -6 & 3 \\ 0 & 0 & 3 & -6 & 3 \end{array}\right]$$

$$\xrightarrow[\frac{1}{3}r_2]{r_3-r_2} \left[\begin{array}{cccc|c} 1 & -2 & -1 & 3 & 1 \\ 0 & 0 & 1 & -2 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right] \xrightarrow[r_1+r_2]{} \left[\begin{array}{cccc|c} 1 & -2 & 0 & 1 & 2 \\ 0 & 0 & 1 & -2 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

The corresponding reduced system of equation

$$\begin{cases} x_1 - 2x_2 \qquad + \quad x_4 & = 2 \\ \qquad\qquad x_3 - 2x_4 & = 1 \\ \qquad\qquad\qquad\qquad 0 & = 0 \end{cases}$$

The leading ones are in columns 1 and 3 here, so the corresponding variables x_1 and x_3 are called leading variables.

Solve for the leading variables in terms of the nonleading variables x_2 and x_4

$$\begin{cases} x_1 - 2x_2 \qquad + \ x_4 & = 2 \\ \qquad\qquad x_3 - 2x_4 & = 1 \\ \qquad\qquad\qquad\qquad 0 & = 0 \end{cases}$$

Set $x_2 = s$, $x_4 = t$ then

$$x_1 - 2s + t = 2 \text{ and } x_3 - 2t = 1$$

The solution is given by

$$\begin{aligned} x_1 &= 2 + 2s - t \\ x_2 &= s \\ x_3 &= 2t + 1 \\ x_4 &= t \end{aligned}$$

where s and t are arbitrary.

Solution of Linear Systems

- To solve a linear system, the augmented matrix is carried to reduced row-echelon form
- the variables corresponding to the leading ones are called **leading variables**
- Because the matrix is in reduced form, each leading variable occurs in exactly one equation, so that equation can be solved to give a formula for the leading variable in terms of the nonleading variables
- call the nonleading variables **"free" variables**
- **label "free" variables** by new variables s, t, ..., called **parameters**
- every variable x_i is given by a formula in terms of the parameters $s, t, \ldots$
- **every choice of these parameters leads to a solution to the system**, and every solution arises in this way

Gaussian Elimination

To solve a system of linear equations proceed as follows:

1. **Carry the augmented matrix to a reduced row-echelon matrix** using elementary row operations.
2. If a row $\begin{bmatrix} 0 & 0 & 0 & \cdots & 0 & 1 \end{bmatrix}$ occurs, the system is inconsistent.
3. **Otherwise**, the system is consistent. To find solution, assign the nonleading variables (if any) as parameters, and use the equations corresponding to the reduced row-echelon matrix to solve for the leading variables in terms of the parameters with **back substitution**

.

Back Substitution Procedure

- The nonleading variables are assigned as parameters.
- The last equation (corresponding to the row-echelon form) is used to solve for the last leading variable in terms of the parameters.
- This last leading variable is then substituted into all the preceding equations.
- Then, the second last equation yields the second last leading variable, which is also substituted back.
- The process continues to give the general solution.

Example

Find a condition on the numbers a, b, and c such that the following system of equations is consistent. When that condition is satisfied, find all solutions (in terms of a, b, and c).

$$\begin{aligned} x_1 + 3x_2 + x_3 &= a \\ -x_1 - 2x_2 + x_3 &= b \\ 3x_1 + 7x_2 - x_3 &= c \end{aligned}$$

Solution

Augmented matrix

$$\left[\begin{array}{ccc|c} 1 & 3 & 1 & a \\ -1 & -2 & 1 & b \\ 3 & 7 & -1 & c \end{array}\right] \xrightarrow[r_2+r_1]{r_3-3r_1} \left[\begin{array}{ccc|c} 1 & 3 & 1 & a \\ 0 & 1 & 2 & a+b \\ 0 & -2 & -4 & c-3a \end{array}\right]$$

$$\xrightarrow[r_3+2r_2]{} \left[\begin{array}{ccc|c} 1 & 3 & 1 & a \\ 0 & 1 & 2 & a+b \\ 0 & 0 & 0 & -a+2b+c \end{array}\right] \xrightarrow[r_1-3r_2]{} \left[\begin{array}{ccc|c} 1 & 0 & -5 & -2a-3b \\ 0 & 1 & 2 & a+b \\ 0 & 0 & 0 & -a+2b+c \end{array}\right]$$

1. If $-a+2b+c \neq 0$, there is no solution
2. If $-a+2b+c = 0$, the system is consistent. The lastest matrix become

$$\left[\begin{array}{ccc|c} 1 & 0 & -5 & -2a-3b \\ 0 & 1 & 2 & a+b \\ 0 & 0 & 0 & 0 \end{array}\right]$$

Set $x_3 = t$ then the general solution is

$$x_1 = 5t - (2a+3b), \quad x_2 = (a+b) - 2t, \quad x_3 = t$$

Three possibilities about solution of a linear system

1. **No solution** (inconsistent) when a row $\begin{bmatrix} 0 & 0 & 0 & \cdots & 0 & 1 \end{bmatrix}$ occurs.
2. **Unique solution** when every variable is a leading variable.
3. **Infinitely many solutions** when the system is consistent and there is at least one nonleading variable

Rank of a Matrix

The rank of matrix A, denote by $rank(A)$, is **the number of leading 1s in any row-echelon matrix** to which A can be carried by row operations.

Rank of matrix can tell you about the consistency/inconsistency of a system equation

Theorem

Consider a system of linear equation with n variables and augmented matrix $\left[A \mid \boldsymbol{b}\right]$ then

1. $rank(\left[A \mid \boldsymbol{b}\right])$ *is either* $rank(A)$ *or* $rank(A)+1$
2. *The system is consistent if* $rank(\left[A \mid \boldsymbol{b}\right]) = rank(A) = r$
   - *if* $r = n$, *the system has unique solution*
   - *if* $r < n$, *the system has infinite many solution. The set of solutions involves exactly* $n - r$ *parameters with respect to the free variables*
3. *The system is inconsistent if* $rank(\left[A \mid \boldsymbol{b}\right]) = rank(A) + 1$

Example

Consider the system equation

$$\begin{cases} x_1 + \ \ x_2 + \ \ x_3 = 1 \\ x_1 + 2x_2 + 3x_3 = 5 \end{cases}$$

We have $A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 5 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 1 \\ 5 \end{bmatrix}.$

The augmented matrix

$$\left[A \mid b\right] = \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 1 \end{array}\right]$$

is carried to row-echelon form $\left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 4 \end{array}\right]$. So $rank(\left[A \mid b\right]) = 2$ The corresponding row echelon form of A is $\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \end{bmatrix}$. Hence $rank(A) = 2$

Because $rank(A) = rank(\left[A \mid b\right]) = r = 2$, the system is consistent, that is it has at least one solution.
Furthermore, the number of variables $n = 3 > r$, the system has many infinite solution. The set of solution is character by $n - r = 3 - 2 = 1$ parameter.
In fact, from the row - echelon form, we obtain an equivalent system

$$\begin{cases} x_1 + x_2 + \ \ x_3 = 1 \\ \qquad\quad x_2 + 2x_3 = 4 \end{cases}$$

which has two leading variables x_1, x_2 and a free variable x_3. Set $x_3 = t$ with $t \in \mathbb{R}$ then

$$\begin{cases} \ \ t + x_1 + x_2 = 1 \\ 2t \qquad\quad + x_2 = 4 \end{cases} \Longrightarrow \begin{cases} x_1 \qquad\quad = t - 3 \\ \qquad\quad x_2 = 4 - 2t \end{cases}$$

The set of solution is $\{(t - 3, 4 - 2t, t) : t \in \mathbb{R}\}$ which is many infinite and depends on a parameter t

Example

Consider the system equation

$$\begin{cases} x_1 + \ \ x_2 + \ \ x_3 = 1 \\ x_1 + 2x_2 + 3x_3 = 3 \\ 2x_1 + 3x_2 + 4x_3 = 5 \end{cases}$$

The reduction procedure leads the augmented matrix to a row-echelon form

$$\left[A \mid \mathbf{b}\right] \to \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 2 \\ 0 & 0 & 0 & 1 \end{array}\right]$$

The corresponding row-echelon form of coefficient matrix A is $\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}$.

Hence $rank(A) = 2 < rank(\left[A \mid \mathbf{b}\right]) = 3$. Then the system has no solution

Example

Consider the system equation

$$\begin{cases} x_1 + \ \ x_2 + \ \ x_3 = 1 \\ x_1 + 2x_2 + 3x_3 = 3 \\ x_1 + \ \ x_2 + 4x_3 = 7 \end{cases}$$

The reduction procedure leads the augmented matrix to a row-echelon form

$$\left[A \mid \mathbf{b}\right] \to \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 2 \\ 0 & 0 & 1 & 2 \end{array}\right]$$

The corresponding row-echelon form of coefficient matrix A is $\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}$.

Hence $rank(A) = rank(\left[A \mid \mathbf{b}\right]) = 3 =$ number of variables. Then the system has unique solution which is $(1, -2, 2)$

Example

Find the conditions on a, b such that the system has no solution, one solution, and infinitely many solutions

$$x - 2y = 1$$
$$ax + by = 5$$

Solution

We have coefficient matrix $A = \begin{bmatrix} 1 & -2 \\ a & b \end{bmatrix}$, *constant matrix* $\boldsymbol{b} = \begin{bmatrix} 1 \\ 5 \end{bmatrix}$ *and the augmented matrix* $\left[A \mid \boldsymbol{b}\right] = \left[\begin{array}{cc|c} 1 & 2 & 1 \\ a & b & 5 \end{array}\right]$.

Need to carry the augmented matrix to a reduced row-echelon matrix

Solution (cont)

We have this system has $m = 2$ linear equation with $n = 2$

$$\left[A \mid \mathbf{b}\right] \xrightarrow[r_2 - ar_1]{} \left[\begin{array}{cc|c} 1 & -2 & 1 \\ 0 & b+2a & 5-a \end{array}\right]$$

A has a same reduced echolon form as $\begin{bmatrix} 1 & -2 \\ 0 & b+2a \end{bmatrix}$. Hence

$$rank(A) = rank\left(\begin{bmatrix} 1 & -2 \\ 0 & b+2a \end{bmatrix}\right)$$

Reduced echolon form of both A and $\left[A \mid \mathbf{b}\right]$ has at least one 1 leading and no more than two 1's leading. So $1 \leq rank(A) \leq 2$ and $1 \leq rank(\left[A \mid \mathbf{b}\right]) \leq 2$.

- The system has no solution if $rank(A) < rank(\left[A \mid \mathbf{b}\right])$. It happens when $rank(A) = 1$ and $rank(\left[A \mid \mathbf{b}\right]) = 2$. It is equivalent to $b + 2a = 0$ and $5 - a \neq 0$.

Solution (cont)

- The system has one solution when $rank(A) = rank(\left[A \mid \mathbf{b}\right]) = n = 2$. It is equivalent to $b + 2a \neq 0$.
- The system has infinite many solutions if

$$rank(A) = rank(\left[A \mid \mathbf{b}\right]) < n$$

The conditions are $b + 2a = 0$ and $5 - a = 0$. So $a = 5$ and $b = -10$

Exercise

Find the conditions on a, b and c such that the system has no solution, one solution, and infinitely many solutions

$$\begin{cases} 3x + \ \ y - \ \ z = a \\ \ \ x - \ \ y + 2z = b \\ 5x + 3y - 4z = c \end{cases}$$

Practice

Three Nissans, two Fords, and four Chevrolets can be rented for \$106 per day. At the same rates two Nissans, four Fords, and three Chevrolets cost \$107 per day, whereas four Nissans, three Fords, and two Chevrolets cost \$102 per day. Find the rental rates for all three kinds of cars.

Practice

A school has three clubs and each student is required to belong to exactly one club. One year the students switched club membership as follows:

- **ClubA.** 40% remain in A, 10% switch to B, 5% switch to C
- **Club B.** 70% remain in B, 20% switch to A, 10% switch to C.
- **Club C**. 60% remain in C, 20% switch to A, 20% switch to B.

If the fraction of the student population in each club is unchanged, find each of these fractions.

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Homogeneous system of equations

Definition

A system of equations in the variables $x_1, x_2, ..., x_n$ is called homogeneous if all the constant terms are zero that is, if each equation of the system has the form

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = 0$$

trivial solution: $x_1 = 0, x_2 = 0, \ldots, , x_n = 0$
Any solution in which at least one variable has a nonzero value is called a **nontrivial solution**

Homogeneous system of equations

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trivial solution: $x_1 = 0, x_2 = 0, \ldots, , x_n = 0$
Any solution in which at least one variable has a nonzero value is called a **nontrivial solution**

A homogeneous system has either uniques (trivial) solution or infinitely may solution

Example

Show that the following homogeneous system has nontrivial solutions.

$$\begin{cases} x_1 - x_2 + 2x_3 - x_4 = 0 \\ 2x_1 + 2x_2 + x_4 = 0 \\ 3x_1 + x_2 + 2x_3 - x_4 = 0 \end{cases}$$

Solution

The reduction of augmented matrix to reduced row- echelon form

$$\left[\begin{array}{cccc|c} 1 & -1 & 2 & -1 & 0 \\ 2 & 2 & 0 & 1 & 0 \\ 3 & 1 & 2 & -1 & 0 \end{array}\right] \xrightarrow[r_2-2r_1]{r_3-3r_1} \left[\begin{array}{cccc|c} 1 & -1 & 2 & -1 & 0 \\ 0 & 4 & -4 & 3 & 0 \\ 0 & 4 & -4 & 2 & 0 \end{array}\right]$$

$$\xrightarrow[\frac{1}{4}r_2]{r_3-r_2} \left[\begin{array}{cccc|c} 1 & -1 & 2 & -1 & 0 \\ 0 & 1 & -1 & \frac{3}{4} & 0 \\ 0 & 0 & 0 & -1 & 0 \end{array}\right] \xrightarrow[r_1+r_2]{-r_3} \left[\begin{array}{cccc|c} 1 & 0 & 1 & \frac{1}{4} & 0 \\ 0 & 1 & -1 & \frac{3}{4} & 0 \\ 0 & 0 & 0 & 1 & 0 \end{array}\right]$$

Solution (cont)

$$\xrightarrow[r_1-\frac{1}{4}r_3]{r_2-\frac{3}{4}r_3} \left[\begin{array}{cccc|c} 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{array}\right]$$

Assign $x_3 = t$ *so* $x_1 = -t$, $x_2 = t, x_4 = 0$. *Taking* $t = 1$, *we get a non trivial solution* $x_1 = -1, x_2 = 1, x_3 = 1, x_4 = 0$.

Solution (cont)

$$\xrightarrow[r_1-\frac{1}{4}r_3]{r_2-\frac{3}{4}r_3} \left[\begin{array}{cccc|c} 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{array}\right]$$

Assign $x_3 = t$ *so* $x_1 = -t$, $x_2 = t, x_4 = 0$. *Taking* $t = 1$, *we get a non trivial solution* $x_1 = -1, x_2 = 1, x_3 = 1, x_4 = 0$.

In general, the solution of this system equation has parametric form

$$\mathbf{x} = \begin{bmatrix} -t \\ t \\ t \\ 0 \end{bmatrix}$$

All component of general solution is multiple of the corresponding component in the above non trivial solution

Equality of rows/columns

As for rows, two columns are regarded as equal if they have the same number of entries and corresponding entries are the same

Sum and scalar of columns

If $\mathbf{x} = \begin{bmatrix} x_1 \\ \dots \\ x_n \end{bmatrix}$ and $\mathbf{y} = \begin{bmatrix} y_1 \\ \dots \\ y_n \end{bmatrix}$ then $\mathbf{x} + \mathbf{y} = \begin{bmatrix} x_1 + y_1 \\ \dots \\ x_n + y_n \end{bmatrix}$ and $k\mathbf{x} = \begin{bmatrix} kx_1 \\ \dots \\ kx_n \end{bmatrix}$

Linear combination

A sum of scalar multiples of several columns is called a linear combination of these columns.

same idea for rows

Example

For example, $s\mathbf{x} + t\mathbf{y}$ is a linear combination of x and y for any choice of numbers s and t.

Example

$\mathbf{x} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$ and $\mathbf{y} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$ then $2\mathbf{x} + 3\mathbf{y} = \begin{bmatrix} 2 \\ 6 \end{bmatrix} + \begin{bmatrix} -3 \\ 3 \end{bmatrix} = \begin{bmatrix} -1 \\ 9 \end{bmatrix}$

Example

Let $\mathbf{x} = \begin{bmatrix} 1 & 0 & 1 \end{bmatrix}$, $\mathbf{y} = \begin{bmatrix} 2 & 1 & 0 \end{bmatrix}$ and $\mathbf{z} = \begin{bmatrix} 3 & 1 & 1 \end{bmatrix}$. If $\mathbf{v} = \begin{bmatrix} 0 & -1 & 2 \end{bmatrix}$ and $\mathbf{w} = \begin{bmatrix} 1 & 1 & 1 \end{bmatrix}$, determine whether $\mathbf{v}$ and $\mathbf{w}$ are linear combination of $\mathbf{x}$, $\mathbf{y}$, $\mathbf{z}$.

Example

Let $\mathbf{x} = \begin{bmatrix} 1 & 0 & 1 \end{bmatrix}$, $\mathbf{y} = \begin{bmatrix} 2 & 1 & 0 \end{bmatrix}$ and $\mathbf{z} = \begin{bmatrix} 3 & 1 & 1 \end{bmatrix}$. If $\mathbf{v} = \begin{bmatrix} 0 & -1 & 2 \end{bmatrix}$ and $\mathbf{w} = \begin{bmatrix} 1 & 1 & 1 \end{bmatrix}$, determine whether **v** and **w** are linear combination of **x, y, z**.

Solution

*For **v**, we need to determine whether there exists number r, s, t such that $\mathbf{v} = r\mathbf{x} + s\mathbf{y} + t\mathbf{z}$, that is*

$$\begin{bmatrix} 0 & -1 & 2 \end{bmatrix} = r \begin{bmatrix} 1 & 0 & 1 \end{bmatrix} + s \begin{bmatrix} 2 & 1 & 0 \end{bmatrix} + t \begin{bmatrix} 3 & 1 & 1 \end{bmatrix}$$

Equating the corresponding entries gives a system of linear equation

$$\begin{aligned} r + 2s + 3t &= 0 \\ s + t &= -1 \\ r + t &= 2 \end{aligned}$$

By gaussian elimination, the solution is $r = 2 - k, s = -1 - k$, and $t = k$ where k is a parameter. Taking $k = 0$, we see that $\mathbf{v} = 2\mathbf{x} - \mathbf{y}$ is a linear combination of $\mathbf{x}$, $\mathbf{y}$, and $\mathbf{z}$.

By gaussian elimination, the solution is $r = 2 - k, s = -1 - k$, and $t = k$ where k is a parameter. Taking $k = 0$, we see that $\mathbf{v} = 2\mathbf{x} - \mathbf{y}$ is a linear combination of **x**, **y**, and **z**.

What's your answer for **w**?

Interest of linear combination

Linear combination is one of the best ways to describe the general solution of homogeneous system of linear equations

If $\mathbf{x}$ and $\mathbf{y}$ are two solutions to a homogeneous system with n variables. Then any linear combination $s\mathbf{x} + t\mathbf{y}$ of these solutions turns out to be again a solution to the system

Any linear combination of solutions to a homogeneous system is again a solution

Basic solution of homogeneous system

General solution of the system of homogeneous linear equations

$$\begin{cases} x_1 - 2x_2 + 3x_3 - 2x_4 = 0 \\ -3x_1 + 6x_2 + \ \ x_3 \qquad\quad = 0 \\ -2x_1 + 4x_2 + 4x_3 - 2x_4 = 0 \end{cases}$$

is $x_1 = 2s + \frac{1}{5}t$, $x_2 = s, x_3 = \frac{3}{5}t, x_4 = t$ and can be written in matrix form

$$\mathbf{x} = \begin{bmatrix} 2s + \frac{1}{5}t \\ s \\ \frac{3}{5}t \qquad t \end{bmatrix} = \begin{bmatrix} 2s \\ s \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} \frac{1}{5}t \\ 0 \\ \frac{3}{5}t \\ t \end{bmatrix} = s\mathbf{x}_1 + t\mathbf{x}_2$$

where $\mathbf{x}_1 = \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \end{bmatrix}$ and $\mathbf{x}_2 = \begin{bmatrix} \frac{1}{5} \\ 0 \\ \frac{3}{5} \\ 1 \end{bmatrix}$ which are called **basic solutions**

Definition (Basic solutions)

The gaussian algorithm systematically produces solutions to any homogeneous linear system, called **basic solutions**, one for every parameter.

Convention

Any nonzero scalar multiple of a basic solution will still be called a basic solution

Theorem

Let A be an $m \times n$ matrix of rank r, and consider the homogeneous system in n variables with A as coefficient matrix. Then:

1. *The system has exactly $n - r$ basic solutions, one for each parameter.*
2. *Every solution is a linear combination of these basic solutions.*

Example

Find basic solutions of the homogeneous system with coefficient matrix A, and express every solution as a linear combination of the basic solutions where

$$A = \begin{bmatrix} 1 & -3 & 0 & 2 & 2 \\ -2 & 6 & 1 & 2 & -5 \\ 3 & -9 & -1 & 0 & 7 \\ -3 & 9 & 2 & 6 & -8 \end{bmatrix}$$

Solution

The reduction of the augmented matrix to reduced row-echelon form is

$$\left[\begin{array}{ccccc|c} 1 & -3 & 0 & 2 & 2 & 0 \\ -2 & 6 & 1 & 2 & -5 & 0 \\ 3 & -9 & -1 & 0 & 7 & 0 \\ -3 & 9 & 2 & 6 & -8 & 0 \end{array}\right] \xrightarrow{\text{reduction procedure}} \left[\begin{array}{ccccc|c} 1 & -3 & 0 & 2 & 2 & 0 \\ 0 & 0 & 1 & 6 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

The general solution is $x_1 = 3r - 2s - 2t$, $x_2 = r$, $x_3 = -6s + t$, $x_4 = s$, $x_5 = t$ where r, s, t are parameters.
In matrix form, this is

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 3r - 2s - 2t \\ r \\ -6s + t \\ s \\ t \end{bmatrix} = r \begin{bmatrix} 3 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + s \begin{bmatrix} -2 \\ 0 \\ -6 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} -2 \\ 0 \\ 1 \\ 0 \\ 1 \end{bmatrix}$$

Solution (cont)

Hence the basic solutions are

$$\mathbf{x}_1 = \begin{bmatrix} 3 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \mathbf{x}_2 = \begin{bmatrix} -2 \\ 0 \\ -6 \\ 1 \\ 0 \end{bmatrix}, \mathbf{x}_3 = \begin{bmatrix} -2 \\ 0 \\ 1 \\ 0 \\ 1 \end{bmatrix}$$

Exercise

Find a set of basic solutions and express the general solution as a linear combination of these basic solutions.

$$\begin{aligned} x_1 + x_2 - x_3 + 2x_4 + x_5 &= 0 \\ x_1 + 2x_2 - x_3 + x_4 + x_5 &= 0 \\ 2x_1 + 3x_2 - x_3 + 2x_4 + x_5 &= 0 \end{aligned}$$

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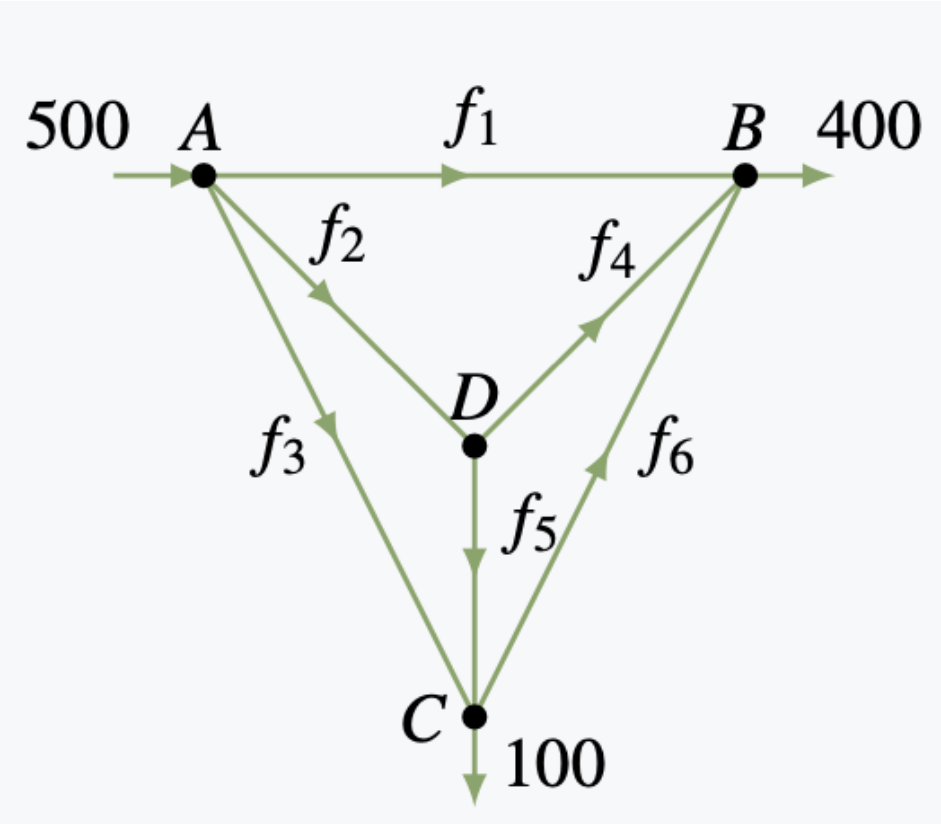
Network Flow Analysis

There are many types of problems that concern a network of conductors along which some sort of flow is observed. Examples of these include an irrigation network and a network of streets or freeways. There are often points in the system at which a net flow either enters or leaves the system. The basic principle (**junction rule**) behind the analysis of such systems is that the total flow into the system must equal the total flow out.

Example

A network of one-way streets is shown in the accompanying diagram. The rate of flow of cars into intersection A is 500 cars per hour, and 400 and 100 cars per hour emerge from B and C, respectively. Find the possible flows along each street.



Solution

Equating the flow in with the flow out at each intersection, we get

$$\begin{aligned} f_1 + f_2 + f_3 &= 500 \\ f_1 + f_4 + f_6 &= 400 \\ f_3 + f_5 - f_6 &= 100 \\ f_2 - f_4 - f_5 &= 0 \end{aligned}$$

The reduction of augmented matrix is

$$\left[\begin{array}{cccccc|c} 1 & 1 & 1 & 0 & 0 & 0 & 500 \\ 1 & 0 & 0 & 1 & 0 & 1 & 400 \\ 0 & 0 & 1 & 0 & 1 & -1 & 100 \\ 0 & 1 & 0 & -1 & -1 & 0 & 0 \end{array}\right] \to \left[\begin{array}{cccccc|c} 1 & 1 & 1 & 0 & 0 & 0 & 500 \\ 0 & 1 & 0 & -1 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & -1 & 100 \\ 0 & 0 & 0 & -1 & -1 & 0 & 0 \end{array}\right]$$

Use f_4, f_5, f_6 as parameters, the general solution is

$$f_1 = 400 - f_4 - f_6, \quad f_2 = f_4 + f_5, \quad f_3 = 100 - f_5 + f_6$$

All possible flows are solutions of the system of equation which satisfy the constraints on flows

$$f_1 \geq 0, f_3 \geq 0, f_4 + f_6 \leq 400, f_5 - f_6 \leq 100$$

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Balance for Chemical Reaction

$$H_2 + O_2 \rightarrow H_2O$$

Individual atoms are neither created nor destroyed, so the number of hydrogen and oxygen atoms going into the reaction must equal the number coming out (in the form of water)

$$2H_2 + O_2 \rightarrow 2H_2O$$

In this case the reaction is said to be balanced

In order to balance the reaction, we need to find positive integers x, y, z such that

$$xH_2 + yO_2 \to zH_2O$$

Equating the number of hydrogen and oxygen atoms on each side give

$$\begin{cases} 2x = 2z \\ 2y = z \end{cases}$$

which can be written as a homogeneous linear system

$$\begin{cases} 2x - 2z = 0 \\ 2y - z = 0 \end{cases}$$

and solved by gaussian elimination. Set $z = t$ then $x = t$ and $y = \frac{1}{2}t$. x, y, z must be positive integers so the smallest value of t is 2. Hence $x = 2, y = 1, z = 2$ and the balance reaction is

$$2H_2 + O_2 \to 2H_2O$$

Exercise

Balance the following reaction for burning octane C_8H_{18} in oxygen O_2:

$$C_8H_{18} + O_2 \rightarrow CO_2 + H_2O$$